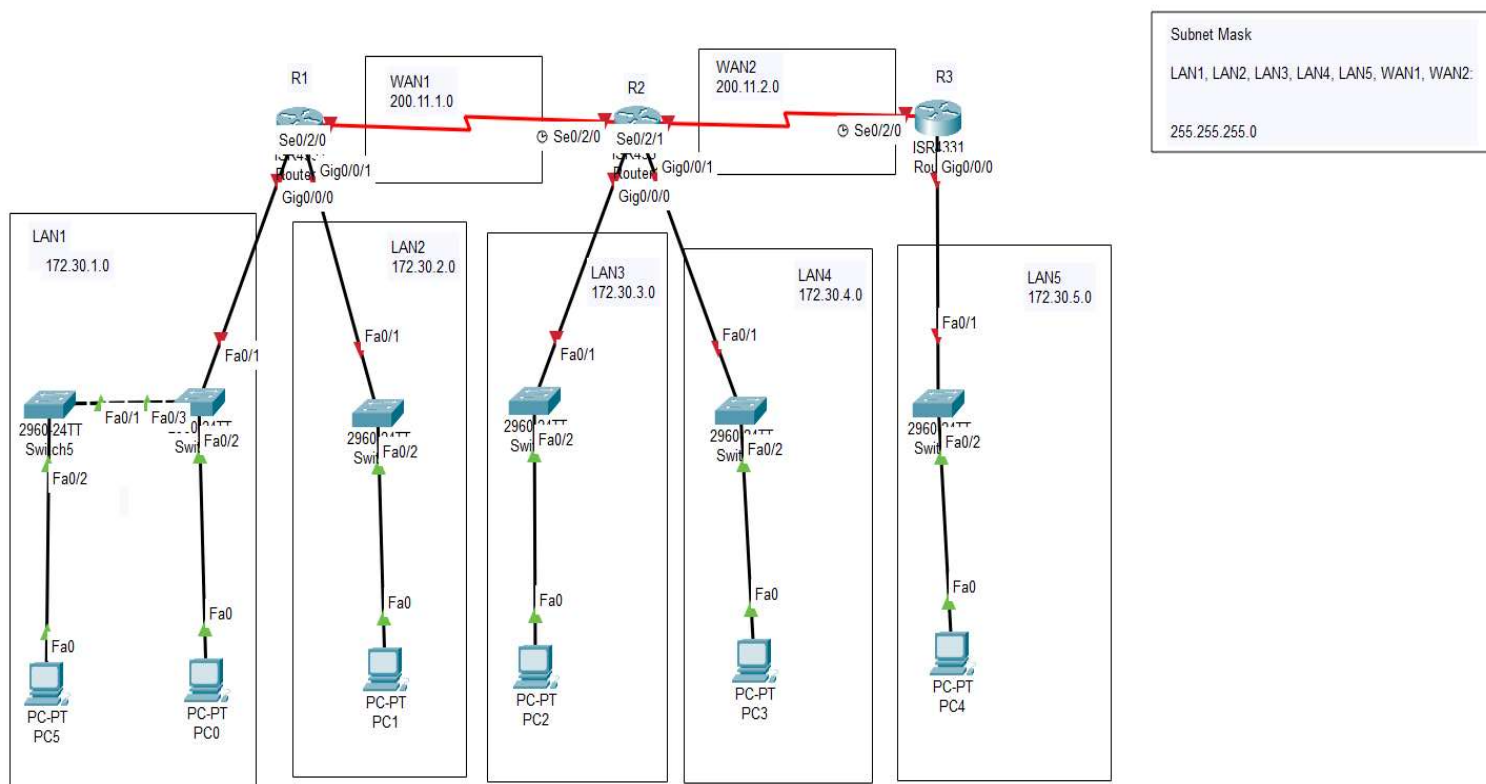


Network Project



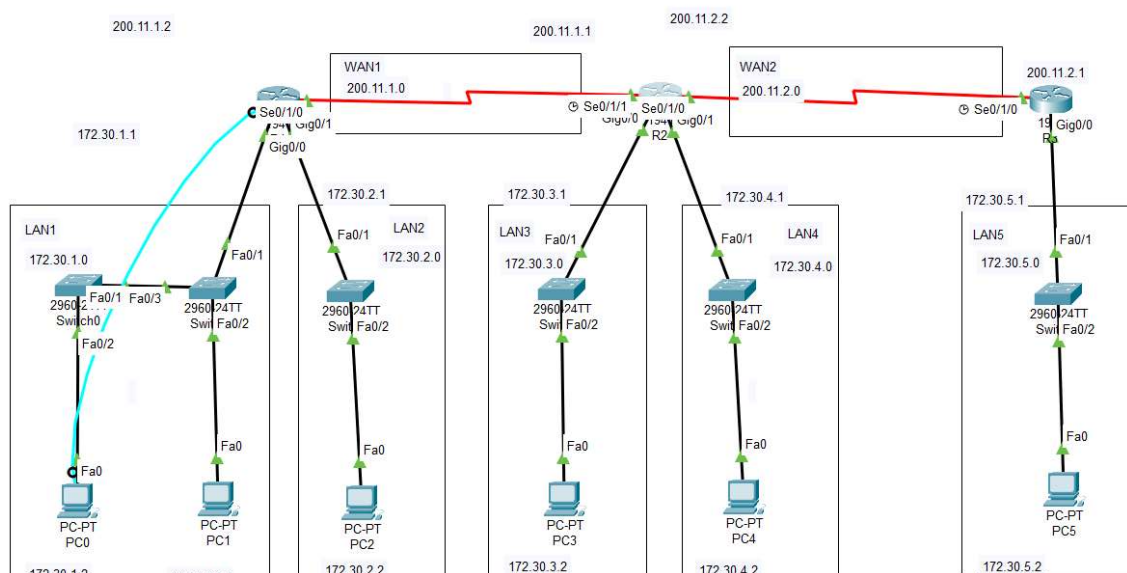
1. Use packet tracer to create the network topology.
2. Configuring device by Adding the IP configuration for each device (PC, Router)
3. Test the ping command between devices in the same Network.
4. Add the routing protocol RIP
5. Test the ping command between device in different Network.

Note:

1. You must take a snapshot for each step and add it in the report.
2. You must send the packet tracer file and the report.

Good luck

• Network Topology Implementation



• IP Address Configuration

Pc:

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.30.1.2

Subnet Mask 255.255.0.0

Default Gateway 172.30.1.1

DNS Server 0.0.0.0

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.30.1.3

Subnet Mask 255.255.0.0

Default Gateway 172.30.1.1

DNS Server 0.0.0.0

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.30.2.2

Subnet Mask 255.255.0.0

Default Gateway 172.30.2.1

DNS Server 0.0.0.0

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.30.3.2

Subnet Mask 255.255.0.0

Default Gateway 172.30.3.1

DNS Server 0.0.0.0

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.30.4.2

Subnet Mask 255.255.0.0

Default Gateway 172.30.4.1

DNS Server 0.0.0.0

PC5

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.30.5.2

Subnet Mask 255.255.0.0

Default Gateway 172.30.5.1

DNS Server 0.0.0.0

Router:

R1

Physical Config CLI Attributes

IOS Command Line

```
!
interface GigabitEthernet0/0
ip address 172.30.1.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 172.30.2.1 255.255.255.0
duplex auto
speed auto
!
interface Serial0/1/0
ip address 200.11.1.2 255.255.255.0
!
```

R2

Physical Config CLI Attributes

IOS Command

```
!
interface GigabitEthernet0/0
ip address 172.30.3.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 172.30.4.1 255.255.255.0
duplex auto
speed auto
!
interface Serial0/1/0
ip address 200.11.2.2 255.255.255.0
!
interface Serial0/1/1
ip address 200.11.1.1 255.255.255.0
clock rate 64000
!
```

R3

Physical Config CLI Attributes

IOS Command

```
!
interface GigabitEthernet0/0
ip address 172.30.5.1 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
ip address 200.11.2.1 255.255.255.0
clock rate 64000
!
```

- Ping Test Within the Same LAN

PC0

Physical
Config
Desktop
Programming
Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.1.3

Pinging 172.30.1.3 with 32 bytes of data:

Reply from 172.30.1.3: bytes=32 time<1ms TTL=128
Reply from 172.30.1.3: bytes=32 time<1ms TTL=128
Reply from 172.30.1.3: bytes=32 time<1ms TTL=128
Reply from 172.30.1.3: bytes=32 time=7ms TTL=128

Ping statistics for 172.30.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 1ms

Ping from PC0 to PC1

PC1

Physical
Config
Desktop
Programming
Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.2.2

Pinging 172.30.2.2 with 32 bytes of data:

Request timed out.
Reply from 172.30.2.2: bytes=32 time<1ms TTL=127
Reply from 172.30.2.2: bytes=32 time<1ms TTL=127
Reply from 172.30.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.2.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

Ping from PC1 to PC2

PC3

Physical
Config
Desktop
Programming
Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.4.2

Pinging 172.30.4.2 with 32 bytes of data:

Request timed out.
Reply from 172.30.4.2: bytes=32 time<1ms TTL=127
Reply from 172.30.4.2: bytes=32 time<1ms TTL=127
Reply from 172.30.4.2: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.4.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

Ping from PC3 to PC4

- **RIP Routing Configuration**

 **R1**

Physical Config CLI Attributes

```
router rip
version 2
network 172.30.0.0
network 200.11.1.0
!
```

 **R2**

Physical Config CLI Attributes

```
router rip
version 2
network 172.30.0.0
network 200.11.1.0
network 200.11.2.0
!
```

 **R3**

Physical Config CLI Attributes

```
router rip
version 2
network 172.30.0.0
network 200.11.2.0
!
```

- Ping Test Between Different Networks

```

PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.5.2

Pinging 172.30.5.2 with 32 bytes of data:

Reply from 172.30.5.2: bytes=32 time=19ms TTL=125
Reply from 172.30.5.2: bytes=32 time=2ms TTL=125
Reply from 172.30.5.2: bytes=32 time=17ms TTL=125
Reply from 172.30.5.2: bytes=32 time=21ms TTL=125

Ping statistics for 172.30.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 21ms, Average = 14ms
    
```

Ping from PC1 to PC5

```

PC3
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.1.2

Pinging 172.30.1.2 with 32 bytes of data:

Reply from 172.30.1.2: bytes=32 time=19ms TTL=126
Reply from 172.30.1.2: bytes=32 time=17ms TTL=126
Reply from 172.30.1.2: bytes=32 time=2ms TTL=126
Reply from 172.30.1.2: bytes=32 time=2ms TTL=126

Ping statistics for 172.30.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 19ms, Average = 10ms
    
```

Ping from PC3 to PC1

```

PC5
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.30.4.2

Pinging 172.30.4.2 with 32 bytes of data:

Reply from 172.30.4.2: bytes=32 time=13ms TTL=126
Reply from 172.30.4.2: bytes=32 time=3ms TTL=124
Reply from 172.30.4.2: bytes=32 time=17ms TTL=124
Reply from 172.30.4.2: bytes=32 time=21ms TTL=124

Ping statistics for 172.30.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 21ms, Average = 13ms
    
```

Ping from PC5 to PC4

Router Access Configuration

- Console Configuration: 102030
- Enable Mode: 1234
- Telnet Configuration: 123